

TONGCHUAN, China

WMO: 539470

Lat: 35.083N

Lon: 109.067E

Elev: 2999

StdP: 13.17

Time Zone: 8.00 (E08)

Period: 82-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.6	16.4	-10.4	3.5	26.1	-5.8	4.5	27.3	16.6	34.4	14.3	35.0	3.7	40	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	89.1	66.9	86.7	66.7	84.1	66.9	73.7	81.5	72.4	79.8	71.1	78.3	7.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.6	130.6	77.3	70.3	124.7	76.6	68.8	118.3	75.5	39.7	81.9	38.4	79.5	37.1	78.4	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
15.7	13.0	11.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	28.2	33.4	41.7	53.1	63.4	70.3	73.5	71.8	61.7	52.8	41.4	30.7
	DBStd	16.83	5.42	5.86	6.75	6.75	5.87	5.14	4.43	4.65	4.84	5.57	7.14	5.84
	HDD50	2345	676	464	267	43	1	0	0	0	31	266	597	
	HDD65	5492	1141	884	722	358	100	13	2	4	118	380	708	1062
	CDD50	3054	0	0	10	137	417	609	727	677	352	117	8	0
	CDD65	727	0	0	0	2	51	172	264	216	20	2	0	0
	CDH74	5463	0	0	0	43	480	1578	1927	1315	109	11	0	0
	CDH80	1622	0	0	0	6	95	534	638	339	10	0	0	0
	WSAvg	5.0	4.6	5.0	5.7	5.7	5.6	5.2	4.8	4.8	4.3	4.8	4.9	4.6
	PrecAvg	23.4	0.2	0.3	0.9	1.7	2.1	2.2	4.5	4.7	3.9	2.0	0.8	0.2
Precipitation	PrecMax	35.0	1.0	1.1	2.4	3.4	5.3	5.5	8.3	13.6	9.1	5.9	2.8	0.7
	PrecMin	13.9	0.0	0.0	0.0	0.2	0.1	0.1	1.7	0.7	0.8	0.3	0.0	0.0
	PrecStd	4.6	0.2	0.3	0.6	0.9	1.5	1.3	1.9	3.0	2.4	1.5	0.6	0.2
	PrecStd	4.6	0.2	0.3	0.6	0.9	1.5	1.3	1.9	3.0	2.4	1.5	0.6	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	48.4	55.2	66.5	79.7	86.9	91.8	92.2	91.9	82.5	78.4	63.9	50.3	
	0.4% MCWB	34.5	40.2	50.5	58.4	62.1	64.1	67.3	68.0	64.0	59.0	49.0	36.9	
	2% DB	44.0	50.2	61.8	73.8	82.5	88.7	88.9	87.1	79.0	71.0	59.3	46.2	
	2% MCWB	32.4	37.3	47.4	56.2	60.2	63.6	68.9	69.2	64.1	57.9	47.7	35.1	
	5% DB	40.6	46.5	57.8	69.8	79.4	85.9	86.7	84.0	75.1	67.0	55.5	42.8	
	5% MCWB	30.7	35.5	45.0	53.8	60.1	63.3	68.8	69.9	62.9	55.9	45.6	33.4	
	10% DB	37.5	43.5	53.7	65.6	76.2	82.8	83.9	81.5	71.6	62.8	51.7	39.4	
	10% MCWB	28.9	33.7	42.5	51.8	59.3	63.0	68.8	69.5	61.8	53.5	44.4	31.3	
	0.4% WB	36.0	41.8	52.4	62.0	67.2	71.2	76.0	75.3	69.7	62.8	52.1	39.3	
	0.4% MCDB	46.3	52.0	62.8	74.7	78.0	83.8	84.0	83.0	79.8	72.8	58.8	46.8	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	33.2	38.9	49.2	58.8	65.4	68.5	74.1	73.7	67.1	59.3	49.6	36.4	
	2% MCDB	42.7	46.9	58.9	70.5	75.9	79.3	81.8	81.6	73.7	67.7	55.8	43.7	
	5% WB	31.3	36.9	46.5	55.9	63.8	67.1	72.7	72.6	65.2	57.2	47.5	34.4	
	5% MCDB	38.8	44.2	54.8	66.2	73.6	77.9	80.2	79.9	71.2	64.4	52.8	40.8	
	10% WB	29.6	35.0	44.2	53.6	62.1	65.7	71.4	71.4	63.5	55.3	45.7	32.1	
	10% MCDB	36.0	42.3	52.3	63.4	70.9	75.9	79.0	77.9	69.2	61.1	50.4	38.3	
Mean Daily Temperature Range	MDBR	15.5	15.8	17.1	17.7	20.0	19.5	15.5	14.6	14.0	15.4	15.1	15.5	
	5% DB	20.0	21.0	24.4	24.3	26.3	26.3	21.8	19.2	21.5	20.5	21.1	20.7	
	5% WB	12.7	12.5	13.8	12.0	11.9	9.6	7.6	7.7	10.4	10.9	13.0	12.6	
	MCDBR	18.7	17.3	21.3	22.9	20.5	19.8	16.4	14.3	16.7	17.3	18.0	19.1	
	MCWBR	12.2	10.6	13.1	13.6	11.2	8.7	8.2	7.5	10.1	10.4	11.7	12.0	
	taub	0.494	0.584	0.605	0.610	0.597	0.635	0.656	0.688	0.630	0.602	0.525	0.460	
	taud	1.748	1.600	1.599	1.628	1.711	1.677	1.685	1.605	1.676	1.666	1.746	1.833	
	Edn at Noon	206	202	214	224	229	220	214	203	206	195	195	207	
All-Sky Solar Radiation	RadAvg	823	987	1326	1582	1709	1754	1635	1428	1134	994	846	768	
	RadStd	83	152	132	143	130	152	200	180	124	151	120	93	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-131	-122

Nomenclature: See separate page